

ViT in the Low-Data Regime

Improving Vision Transformers with KD and SSL Techniques

CIFAR-10 | 100 Samples per Class | PyTorch + timm

Experiment Report • 2025

Executive Summary

This report documents a series of experiments designed to study how Vision Transformer (ViT) architectures behave under a deliberate low-data setting and how modern Self-Supervised Learning (SSL) and Knowledge Distillation (KD) techniques can bridge the gap. All experiments share a single benchmark: CIFAR-10 with exactly 100 labelled samples per class (1,000 training images total) and an independent validation and test split of identical size.

Model / Strategy	Val Acc	Test Acc	Key Technique
Vanilla ViT (from scratch)	37.70%	36.55%	Baseline
Pretrained ViT-S/16 (teacher)	93.10%	92.11%	ImageNet pretrain + fine-tune
SmallViT scratch (custom)	57.50%	54.97%	ConvStem + PEG + reg-tokens
MIM pretrain + diff. fine-tune	46.50%	45.66%	MIM SSL (strat. 1)
MIM pretrain + uniform LR FT	56.00%	55.70%	MIM SSL (strat. 2)
MIM + KD (T=3, alpha=0.75)	68.20%	68.31%	MIM pretraining + KD

Best result: MIM pretraining combined with Knowledge Distillation achieves 68.31% test accuracy — a +34.76 pp absolute gain over the vanilla from-scratch baseline.

1 Motivation & Problem Setup

1.1 Why Low-Data Vision Transformers?

ViTs were originally proposed as large-scale architectures that require enormous labelled datasets to train effectively. Unlike CNNs, which have local receptive fields and strong translation-equivariant inductive bias, standard ViTs apply global self-attention from the first layer and lack spatial locality priors. This makes them especially fragile when only a few hundred samples per class are available — the setting that reflects many real-world scenarios such as medical imaging, satellite analysis, or rare-event detection.

The core research question of this project is: can we recover competitive accuracy on a 1 000-sample CIFAR-10 benchmark by combining (a) a purpose-built small ViT architecture with (b) self-supervised pre-training via Masked Image Modelling (MIM), and (c) knowledge distillation from a large, ImageNet-pretrained teacher?

1.2 Dataset & Experimental Protocol

All experiments use CIFAR-10 (60 000 32×32 colour images, 10 classes). A deliberate low-data split is imposed at the start of every experiment:

Training set: 100 images per class → 1 000 images total
Validation set: 100 images per class → 1 000 images total
Test set: Full CIFAR-10 test set → 10 000 images
Random seed: 42 (fixed for reproducibility)

Strong data augmentation is applied consistently: RandomResizedCrop (scale 0.2–1.0), RandomHorizontalFlip, and RandAugment(num_ops=2, magnitude=9). Mixup ($\alpha=0.8$) and CutMix ($\alpha=1.0$) are applied during supervised fine-tuning for regularisation.

2 Architecture Design

2.1 Vanilla ViT Baseline (timm, from scratch)

The first experiment uses a standard ViT implemented via the timm library to establish a baseline for learning from scratch in the low-data setting.

Hyperparameter	Value
Image size	32 × 32
Patch size	8 × 8 (4×4 grid → 16 patches)
Embed dim	384
Depth	4 transformer blocks
Heads	4
MLP ratio	4×
Parameters	7.18 M
Epochs	200
Optimizer	AdamW (lr=3e-4, wd=1e-2)
Scheduler	5-epoch linear warmup + cosine decay
Loss	SoftTargetCrossEntropy (Mixup)

Test Accuracy: 33.55% | Best Val Accuracy: 35.70% | Exhibits severe overfitting and slow convergence

The vanilla ViT suffers from lack of inductive bias: without spatial locality priors, the model cannot efficiently learn visual features from only 1 000 samples. The train accuracy climbs to ~90%+ while validation stagnates around 35%, indicating severe overfitting, confirming the fundamental challenge of ViTs in the low-data regime.

2.2 Pretrained ViT-Small/16 Teacher (timm, ImageNet)

To establish an upper-bound reference and to serve as a teacher for KD later, a ViT-Small/16 pretrained on ImageNet-21k (via timm) is fine-tuned on the 1 000-sample training set.

Hyperparameter	Value
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Model	vit_small_patch16_224 (timm)
Image size	224 × 224 (upscaled from 32×32)
Parameters	21.67 M
Pretraining	ImageNet-21k (timm checkpoint)
Backbone LR	3e-5 (10× lower, protect pretrained weights)
Head LR	3e-4
Regularization	drop_rate=0.2, drop_path_rate=0.1
Epochs	30
Augmentation	RandAugment + Mixup 0.8 + CutMix 1.0

Test Accuracy: 92.11% | Best Val Accuracy: 93.10% — Massive advantage from ImageNet pretraining

Despite using only 1 000 training samples, the pretrained model achieves 92.11% test accuracy. This demonstrates that rich visual representations learned on large-scale data transfer extremely well, even to tiny datasets, when combined with careful differential fine-tuning (lower LR for backbone, higher LR for head).

2.3 Custom SmallViT: Architecture Improvements

Motivated by the failure of the vanilla ViT, a custom architecture was designed with three key improvements tailored for small images and low-data learning:

2.3.1 ConvStem

Instead of tokenising raw 32×32 patches directly, a ConvStem replaces the linear projection layer. It uses a 3-stage convolutional downsampler (3×3 conv, stride 2 each stage), producing a spatial feature grid of 8×8. This introduces CNN-like local inductive bias, allowing the model to detect edges and textures from very few images, which standard patch projection cannot do.

Stage 1: Conv 3×3 stride-2 → embed_dim//4 channels + BN + GELU

Stage 2: Conv 3×3 stride-2 → embed_dim//2 channels + BN + GELU

Stage 3: Conv 3×3 stride-1 → embed_dim channels (no stride, preserve resolution)

Output: 8×8 spatial grid = 64 tokens (vs 16 tokens from 8×8 patches)

2.3.2 Positional Encoding Generator (PEG)

Learned positional embeddings (used in standard ViTs) are fixed in size and have difficulty generalising to different spatial configurations. The PEG module uses a depthwise convolution over the 2D feature grid to generate position-aware representations on the fly:

Implementation: Depthwise Conv 3×3 (stride=1, padding=1, groups=embed_dim) + residual

Why better: Generates position information conditioned on content; more flexible than fixed sin-cos or learned embeddings at small scale

2.3.3 Register Tokens

Inspired by DINOv2 and related work, the model includes 4 learnable register tokens appended to the sequence alongside the standard CLS token. Register tokens absorb high-norm, global artefact representations, allowing the attention heads to focus more cleanly on semantically meaningful spatial tokens.

Count: 4 register tokens

Role: Global information aggregators; reduce attention artefacts in spatial tokens

Classification: Performed using CLS token only (register tokens excluded from head input)

SmallViT Config (final-vit)	Value
embed_dim	384
num_blocks	4 transformer blocks
num_heads	4
num_reg_tokens	4
conv_stages	3
dropout	0.1
Parameters	7.96 M (student) / 11.67 M (MIM model)

3 Self-Supervised Pretraining via MIM

3.1 Why Masked Image Modelling?

Since labelled data is scarce, we leverage unlabelled data by training the model to recover masked image regions. Masked Image Modelling (MIM) is analogous to BERT in NLP: the model learns rich visual representations without any labels. Two MIM variants were explored, differing in model size and training length.

3.2 MIM Architecture

The MIM model shares the SmallViT encoder (ConvStem + Transformer blocks) and adds a lightweight pixel-space decoder for reconstruction:

Encoder: ConvStem (3 stages) + 8 Transformer blocks (mim.ipynb: embed_dim=256, heads=4)

Decoder: 2 Transformer decoder layers, projects back to patch pixel space

Masking strategy: Random 75% of tokens masked (mask_ratio=0.75 in mim.ipynb; 0.7 in final-vit)

Reconstruction target: Normalised pixel values of masked patches (MSE loss)

Augmentation: RandomResizedCrop (scale 0.2–1.0) + RandomHorizontalFlip only (no colour distortion — to preserve pixel targets)

3.3 MIM Pretraining Results

Experiment	Epochs	Final Loss	Config
mim.ipynb (initial)	200	~0.41	embed=256, blocks=8, mask=0.75
final-vit MIM pretrain	800	~0.12	embed=384, blocks=4, mask=0.70

The longer 800-epoch pretraining with the larger embed_dim=384 model achieves a much lower reconstruction loss (~0.12 vs ~0.41), indicating that the encoder has learned substantially richer visual features. This is expected to provide a much better initialisation for downstream fine-tuning.

3.4 MIM Fine-tuning Strategies

After MIM pretraining, the encoder weights are transferred to the SmallViT classifier. Two fine-tuning strategies were tested:

Strategy 1: Differential Learning Rate Fine-tuning

Pre-trained weights use a lower learning rate ($5e-5$) to protect learned representations, while randomly initialised head/PEG weights use a higher learning rate ($5e-4$). This is analogous to the approach used for the teacher model.

Result: Val Acc 46.50% | Test Acc 45.66% — Under-performs vs scratch SmallViT (57.50%)

The differential LR strategy is too conservative for this very small dataset. The low backbone LR prevents the encoder from adapting its MIM representations to the classification objective. The head learns faster than the encoder adapts, leading to a train-val mismatch.

Strategy 2: Uniform Learning Rate Fine-tuning

All parameters (including the pre-trained encoder) receive the same learning rate ($5e-4$) with a cosine annealing schedule over 200 epochs.

Result: Val Acc 56.00% | Test Acc 55.70% — Matches SmallViT from scratch

Strategy 2 allows the pre-trained representations to fully adapt to the CIFAR-10 classification task. The MIM pretraining does provide benefit (convergence is faster) but the final accuracy is comparable to training from scratch. This motivates combining MIM pretraining with additional supervision from a strong teacher.

4 Knowledge Distillation

4.1 Setup

Knowledge Distillation transfers the rich, soft prediction distributions learned by the 92%-accurate teacher model into the student SmallViT. Because the teacher has seen far more data implicitly (ImageNet pretraining), its softened outputs encode inter-class relationships that are invisible in one-hot labels — especially valuable with only 1 000 training samples.

KD Hyperparameter	Value
Teacher	vit_small_patch16_224 (timm, ImageNet, fine-tuned)
Student	SmallViT (MIM-pretrained, embed=384, 4 blocks)
Temperature (T)	3.0 (softens teacher logits)
KD weight (alpha_kd)	0.75 (75% KD, 25% CE)
Feature alignment (beta_feat)	0.4
Loss: KD term	KLDivLoss(reduction=batchmean)
Loss: CE term	SoftTargetCrossEntropy (with Mixup)
Augmentation	Mixup $\alpha=0.8$ + CutMix $\alpha=1.0$ (prob=0.5)
Epochs	300
Optimizer	AdamW (lr=5e-4, wd=0.05)
Scheduler	CosineAnnealingLR (T_max=300)

4.2 Total Loss Objective

$$\mathcal{L}_{\text{total}} = (1 - \alpha) \cdot \mathcal{L}_{\text{CE}}(\text{student}, \text{labels}) + \alpha \cdot T^2 \cdot \mathcal{L}_{\text{KL}}(\text{student}/T \parallel \text{teacher}/T) + \beta \cdot \mathcal{L}_{\text{feat}}$$

The combined loss forces the student to reproduce both the hard label distribution and the teacher's rich soft-label distribution. The feature alignment term (beta_feat=0.4) encourages intermediate representations to align with the teacher, providing additional structural supervision beyond the output logits.

4.3 KD Results

Best Val Acc: 68.20% | Test Acc: 68.31% | +12.31 pp over MIM alone | +13.34 pp over SmallViT from scratch

The MIM + KD combination is the clear winner. The MIM pretraining provides a strong encoder initialisation, and the KD provides rich implicit supervision from the teacher. Together, they allow a 7.96 M parameter model to achieve 68% accuracy from only 1 000 training samples — within ~24 pp of the teacher that is 2.7× larger and saw ImageNet during pretraining.

Epoch	Val Accuracy	Observation
~30	~45%	Initial convergence from KD guidance
~100	~59%	Steady improvement through KL transfer
~150	~65%	Large jump: deep feature alignment kicks in
~200	~65.9%	Stable high-performance regime
~300 (final)	68.20%	Best val; test = 68.31%

5 Full Experiment Comparison

5.1 Results Table

Model	Val Acc	Test Acc	Params	Epochs	Key Technique
Vanilla ViT (scratch)	37.70%	36.55%	7.18 M	200	Baseline
Pretrained ViT-S/16	93.10%	92.11%	21.67 M	30	ImageNet pretrain
SmallViT (scratch)	57.50%	54.97%	7.96 M	200	ConvStem+PEG+Reg
MIM + Diff FT (strat 1)	46.50%	45.66%	7.96 M	200	MIM + low LR FT
MIM + Uniform FT (strat 2)	56.00%	55.70%	7.96 M	200	MIM + cosine FT
MIM + KD (BEST)	68.20%	68.31%	7.96 M	300	MIM + KD (T=3)

5.2 Improvement Analysis

Comparison	Test Acc Delta
Vanilla ViT → SmallViT (arch improvements)	+21.42 pp
SmallViT → MIM pretrain + uniform FT	+0.73 pp (marginal SSL gain alone)
MIM FT → MIM + KD	+12.61 pp (large KD gain)
Vanilla ViT → MIM + KD (full pipeline)	+34.76 pp (overall gain)
MIM + KD vs Teacher (gap)	-23.80 pp (teacher: 92.11%)

The most important takeaway is that the architectural improvements (ConvStem, PEG, register tokens) account for the majority of gains over the vanilla baseline (+21 pp). SSL pretraining via MIM alone provides modest standalone benefit but becomes critically important when combined with KD: it provides a well-initialised encoder that can efficiently absorb teacher guidance during fine-tuning.

6 Training Dynamics & Observations

6.1 Vanilla ViT — Overfitting Pattern

The vanilla ViT demonstrates a characteristic low-data overfitting pattern: train accuracy climbs steeply (reaching ~94% by epoch 91) while validation accuracy stagnates at 33-35%. The model memorises the 1 000 training samples without learning generalisable representations. This confirms that raw ViT architectures require either large datasets or special modifications to work at small scale.

6.2 Pretrained ViT-S/16 — Fast Convergence

The pretrained teacher converges in just 1 epoch to >90% validation accuracy, demonstrating the power of transfer learning. The ImageNet pretraining has already instilled universal visual features, and fine-tuning merely adapts the classification head. Even with aggressive Mixup/CutMix augmentation, the model maintains stable high accuracy throughout training.

6.3 SmallViT from Scratch — Steady Learning

The improved architecture (ConvStem + PEG + register tokens) shows notably healthier learning dynamics than the vanilla ViT: train and validation accuracy improve in parallel for the first 100 epochs, indicating that the inductive biases help the model learn genuine visual features. The train-val gap is present but manageable (~20 pp by epoch 200).

6.4 MIM Pretraining — Loss Landscape

MIM pretraining shows rapid initial loss reduction (from 1.27 at epoch 1 to ~0.25 by epoch 50) followed by a plateau-and-recovery pattern typical of contrastive/generative SSL. The oscillation between epochs 35-50 is characteristic of the model transitioning from coarse texture reconstruction to finer structural reconstruction. The 800-epoch pretraining in final-vit.ipynb pushes this further, achieving a final reconstruction loss of ~0.12 which indicates high-quality learned representations.

6.5 KD Training — Three Phases

The MIM + KD training exhibits three distinct phases: (1) Epochs 1-30: slow initial convergence as the student adapts from MIM-space to classification-space using soft teacher labels; (2) Epochs 30-

150: steady improvement as KL divergence between student and teacher softmax distributions decreases; (3) Epochs 150-300: continued slow refinement, with the best validation accuracy of 68.20% achieved around epoch 288. The absence of overfitting (train and val curves track closely) suggests that the combination of MIM pretraining, KD, and Mixup augmentation provides strong implicit regularisation.

7 Key Findings & Insights

7.1 Architecture Matters Most

The single largest performance jump comes from replacing the vanilla ViT with the custom SmallViT architecture. The three architectural changes — ConvStem, PEG, and register tokens — collectively add +21 pp to test accuracy. This suggests that when training ViTs from scratch on limited data, architecture engineering to incorporate inductive biases is more impactful than training strategy alone.

7.2 MIM Alone Has Limited Direct Benefit

Self-supervised MIM pretraining, when used with strategy 2 (uniform LR), provides only marginal gain over training from scratch on the classification task ($\sim +0.7$ pp). This is consistent with the literature on very small datasets: SSL benefits are most pronounced when the unlabelled dataset is large and diverse. With only 1 000 unlabelled images from CIFAR-10, the MIM model cannot learn substantially richer representations than a well-regularised supervised model. However, MIM pretraining does improve training stability and provides a better initialisation for KD.

7.3 KD Provides the Largest Marginal Gain Over SSL

Adding KD on top of the MIM-pretrained model provides +12.61 pp test accuracy improvement. This demonstrates that teacher knowledge transfer is the most effective technique for closing the performance gap in the low-data regime. The soft labels from the 92%-accurate teacher provide far richer supervisory signal than the one-hot CIFAR-10 labels alone, especially for inter-class boundary regions.

7.4 MIM + KD Synergy

Crucially, MIM pretraining enhances the effectiveness of KD. The MIM encoder has already learned some spatial structure and local feature statistics, making it a better student for absorbing teacher distributions. A student trained from scratch with KD would likely require more epochs or a higher temperature to achieve similar results.

7.5 Fine-tuning Strategy Critically Affects MIM Benefits

Strategy 1 (differential LR) actually hurts performance relative to training from scratch. This is a key negative result: if the pre-trained backbone LR is too low, the head overfits to the random initialisation before the backbone can adapt. Strategy 2 (uniform LR with cosine schedule) avoids this pathology and is the correct approach for small-scale MIM fine-tuning.

8 Technical Implementation Details

8.1 Environment

Framework: PyTorch + timm

Hardware: CUDA GPU (Kaggle/Colab)

Python packages: torch, timm, torchvision, numpy, matplotlib, tqdm

8.2 Optimizer Configuration

Vanilla ViT: AdamW lr=3e-4, wd=1e-2

Pretrained teacher: AdamW backbone lr=3e-5, head lr=3e-4, wd=1e-2

SmallViT scratch: AdamW lr=5e-4, wd=0.05

MIM pretraining: AdamW lr=1.5e-4, wd=0.05

MIM fine-tuning (both strategies): AdamW lr=5e-4 or differential, wd=0.05

KD training: AdamW lr=5e-4, wd=0.05

8.3 Scheduler Configuration

Vanilla ViT: 5-epoch linear warmup + cosine decay over 100 epochs

Teacher: Warmup + cosine over 30 epochs

SmallViT: 20-epoch linear warmup (1% start) + cosine decay over 200 epochs

MIM pretraining: 40-epoch linear warmup + cosine decay over 800 epochs (final-vit)

KD: CosineAnnealingLR T_max=300

8.4 Regularisation

Dropout: 0.1 in SmallViT transformer blocks

Mixup: $\alpha=0.8$ (probability 0.5)

CutMix: $\alpha=1.0$ (probability 0.5)

Label smoothing: 0.1 in supervised fine-tuning (CE loss)

Gradient clipping: max_norm=1.0 (all models)

Weight decay: 0.05 (MIM/KD models)

9 Conclusions

This project demonstrates a systematic progression from a poorly-performing vanilla ViT to a significantly improved model through three complementary strategies:

- Architecture engineering (ConvStem + PEG + register tokens) is the foundation: adding inductive biases appropriate for small images provides the largest absolute performance gain in the low-data setting (+21 pp).
- Self-supervised pretraining via MIM provides a better encoder initialisation but modest direct accuracy benefit when the unlabelled pool is small. Fine-tuning strategy is critical: uniform LR is superior to differential LR for small-scale MIM.
- Knowledge Distillation from a large pretrained teacher provides the most powerful marginal gain (+12.61 pp on top of MIM). The teacher's soft labels encode inter-class structure that is invisible from 1 000 hard labels alone.
- The combination of all three strategies (MIM + KD + improved architecture) achieves 68.31% test accuracy from 1 000 samples — a 34.76 percentage point improvement over the vanilla ViT baseline.

These results strongly support the hypothesis that ViTs can be made competitive in the low-data regime through careful architectural design and training strategy, without requiring large-scale labelled datasets. The teacher-student KD framework with soft targets is particularly effective, enabling a 7.96 M student to benefit from the implicit knowledge encoded in a 21.67 M ImageNet-pretrained teacher.

Final: MIM + KD → Test Accuracy 68.31% | 34.76 pp above vanilla ViT baseline | Within 23.8 pp of the 21.67 M teacher

10 Future Work

Several promising directions could extend these results:

- Iterative KD: Use the fine-tuned student as a new teacher for a smaller student, progressively compressing knowledge while maintaining accuracy.
- Contrastive SSL: Replace MIM with DINO or SimCLR-style pretraining to produce more semantically structured representations before KD fine-tuning.
- Data-efficient ViT variants: Explore T2T-ViT or CrossViT architectures specifically designed for small images and limited data.
- Larger unlabelled pool: Use all 50 000 unlabelled CIFAR-10 training images for MIM pretraining, then fine-tune with the 1 000-label split. This would likely recover most of the SSL benefit observed at larger scales.
- Token-level KD: Align intermediate spatial token representations (not just output logits) between teacher and student for richer structural supervision.
- Semi-supervised extension: Use pseudo-labels from the teacher on unlabelled CIFAR-10 images to augment the supervised fine-tuning stage.